Final Project Information

Irma Zakia
ET4244 Optimization for Telecommunications
School of Electrical Engineering and Informatics
Institut Teknologi Bandung

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Final Project Assignment

1. Identify the background/motivation
2. Explain the optimization problem formulation
3. If the problem is nonconvex NLP:
   a) analyze what approach do they use to solve the nonconvexity, e.g.
      - convert nonconvex problem to convex problem, ideally or approximately
      - use method that guarantees local optimality, e.g. Successive Convex Approximation (SCA), Block Coordinate Descent (BCD), etc.

   b) Solve it *globally* by using at least two *global* solvers like Counne, ANTIGONE, Baron, SCIP, etc. Use less number of variables if computationally more feasible, e.g. less number of users / nodes.
   c) Solve it *locally* using at least two frameworks, e.g. Pyomo, GAMS, AMPL. For each framework: use at least two *local* NLP solvers, e.g. IPOPT, MINOS.
   d) Instead of 3c), you can re-implement the local method proposed in the chosen paper, e.g. SCA, BCD, etc.
   e) Compare the global versus local implementation, i.e. the method in 3c) or 3d) compare to 3b). Comparison in terms of Big-O complexity, computation time, number of iterations to converge, optimal solution and optimal objective

Final Project Assignment

4. Else if the problem is MINLP/MILP:
   a) analyze what approach do they use to solve the nonconvexity, e.g.
      - convert nonconvex problem to convex problem, ideally or approximately
      - use method that guarantees local optimality, e.g. Successive Convex Approximation (SCA), Block Coordinate Descent (BCD), etc.
   b) Solve it *globally* by using at least two global solvers like Counne, ANTIGONE, Baron, SCIP, etc. Use less number of variables if computationally more feasible, e.g. less number of users / nodes.
   c) Solve it *locally* $\rightarrow$ see the slide *Solving MINLP with Nonconvex Function Locally* **or**
      - You must implement Branch and Bound for the integer variables manually
      - then use local NLP solvers for the relaxed continuous variables. You must use these two local solvers: IPOPT and MINOS.
   d) Instead of 4c), you can re-implement the local method proposed in the chosen paper, e.g. SCA, BCD, etc.
   e) Compare the global versus local implementation, i.e. the method in 4c) or 4d) compare to 4b). Comparison in terms of Big-O complexity, computation time, number of iterations to converge, optimal solution and optimal objective

Final Project Assignment

5. Else if the problem is convex programming:
   a) analyze why the function is convex and provide figure to illustrate the convexity using a simpler version of the function, e.g. in 3D
   b) analyze why the constraints are convex and provide figure to illustrate the convexity with a simpler version of the constraints, e.g. in 3D
   c) Re-implement manually by selecting at least two known local methods for constrained problem: Primal-Dual Method, IPM-PD, IPM-LB, ALM, QPM
   d) Compare the methods chosen in 5c) in terms of Big-O complexity, computation time, number of iterations to converge, optimal solution and optimal objective
   e) Compare the results of 5d) with the results using at least two frameworks, e.g. CVXPY, Pyomo, GAMS, AMPL. For each framework, pick at least two appropriate convex solvers, e.g. IPOPT, MINOS, CPLEX, MOSEK, etc.

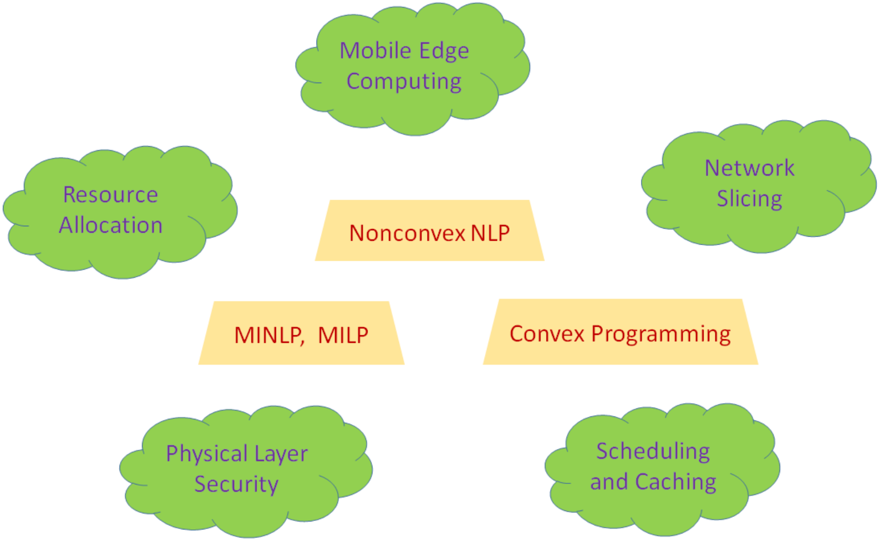
Final Project Format, Deadline, and Presentation

- Individual or a group of maximum 2 students
- Choose one paper from the list. Otherwise, students can propose their own paper, but my approval is required. Paper selection until Monday, May 11, 2026.
- Work on the project based on the slide *Final Project Assignment*
- Project report is in Presentation Format
- Report submission on Edunex. Deadline : Thursday, June 11, 2026 (based on Final Exam schedule)
- Each individual or group deliver presentation on Thursday, June 11, 2026.
- Presentation format:
  - 10 minutes presentation
  - 7-10 minutes Q/A
- Optional :
  - Shorter-version PPT for presentation
  - Longer-version PPT for submission via Edunex

Grading Aspects

1. All questions in slide ”Final Project Assignment” shall be answered exactly as specified.
2. Accuracy of simulation results.
3. Ability in analyzing simulation results.
4. Ability in comparing between different methods and solvers.
5. Ability in explaining project outcomes.

Selected Papers List



Selected Papers List

- List of selected papers available at https://drive.google.com/drive/folders/1f-BvtUdbuHh2AQWyjUfFpgxntVPlxKMO?usp=sharing
- Each student/group is required to select a different paper.
- Confirm your paper selection by filling this sheet https://docs.google.com/spreadsheets/d/1iqYUIcuLhkn-R_kt8zAbueMZ9xp264J_VYugVWMRdsg/edit?usp=sharing
- Deadline: Monday, May 11, 2026